Dataset Selection and Justification

After discussing three possible datasets suggested by different team members, we decided to proceed with the Telco Customer Churn dataset.

We selected this dataset for various practical reasons. First, it contains a good balance of categorical and numerical features, which allowed us to explore different preprocessing techniques such as encoding and scaling. Second, the dataset contains 7043 instances (data points) and 20 features (excluding target feature), which is large enough to train machine learning models effectively while being manageable for analysis.

Finally, the target variable Churn, which indicates whether a customer left the company plays a major role in analysis of customer retention. Customer retention is one of the most important metrics for any organization, being able to predict churn has real business value, as companies can take preventive action to retain customers.

Dataset Description and Target Feature

Selected dataset includes customer demographic information, account details, control type, service subscriptions, billing and payment methods. Target feature is Churn, which has two possible values:

- Yes (Customer Left)
- No (Customer Stayed)

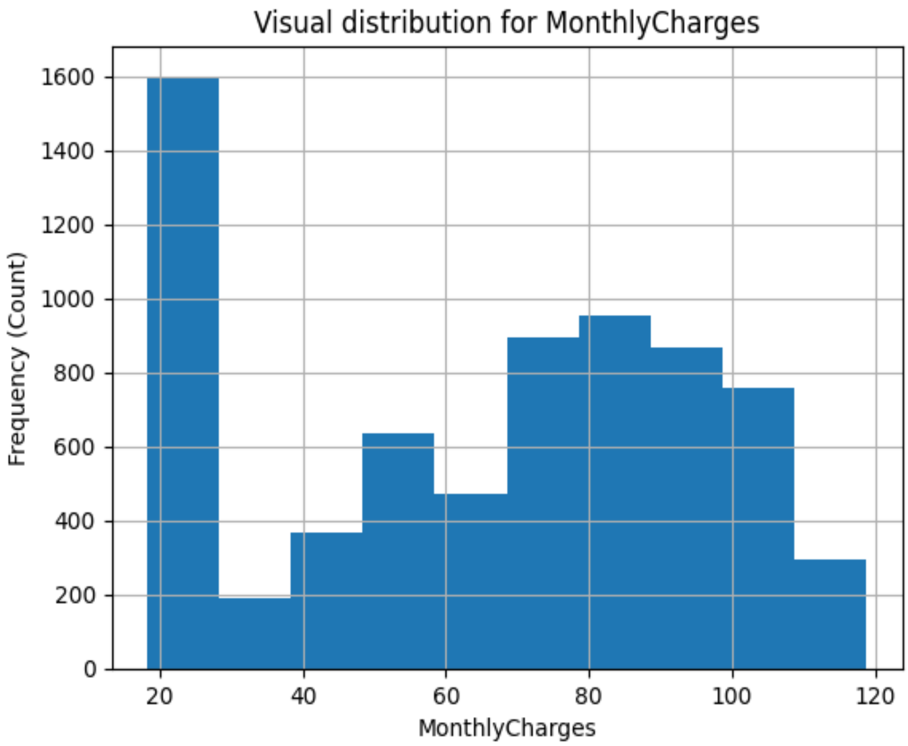
This makes the problem a binary classification problem, where the goal is to predict whether a customer will churn or not. The numerical features include:

- Tenure (number of months the customer has been with the company)
- Monthly Charges (amount charged per month)
- Total Charges (total amount charges over the customer’s time with the organisation)

The remaining features are categorical and represent customer characteristics and service-related information. Most categorical features have low cardinality (two or three categories), which makes them suitable for one-hot encoding during preprocessing.

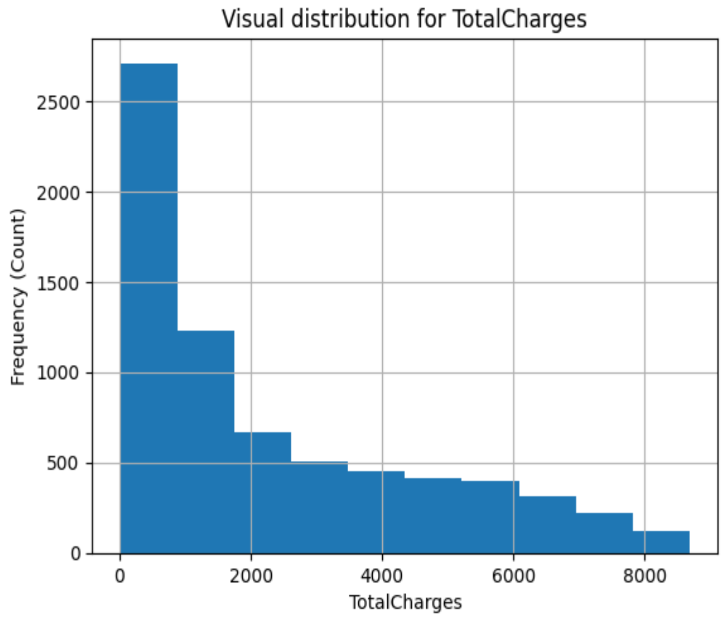
Analysis of Numerical Features

MonthlyCharges:



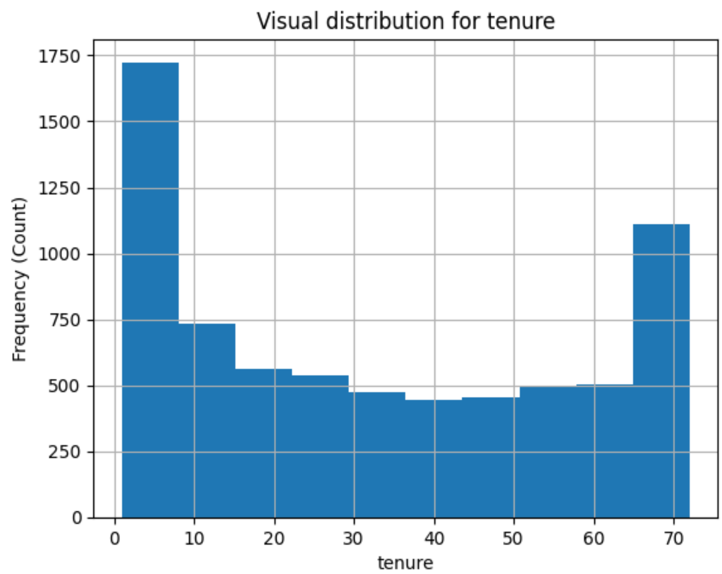
The histogram for the MonthlyCharges feature mostly resembles a bell-shaped curve, although there is a noticeable concentration of customers with lower monthly charges. The X-axis represents the monthly charge amount, and the Y-axis represents the number of customers. This suggests that many customers subscribe to lower-cost plans.

TotalCharges:



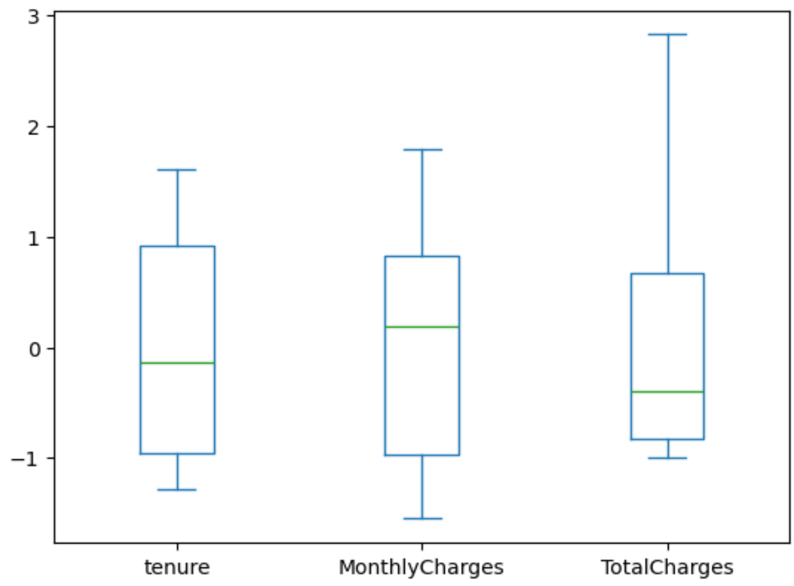
The histogram for TotalCharges appears to be skewed right with an inverse relationship to target feature. The X-axis represents the total charge per customer, and the Y-axis represents the number of customers corresponding to that amount. We observe that as total charges increase, the number of customers decreases. This inverse relationship suggests that fewer customers accumulate very high total charges.

Tenure:



The histogram for Tenure appears bimodal. The X-axis represents the number of months a customer has been with Telco, and the Y-axis represents the number of customers. The bimodal graph suggests two major groups, which indicates customers who recently joined and customers who have been with the company for a long time.

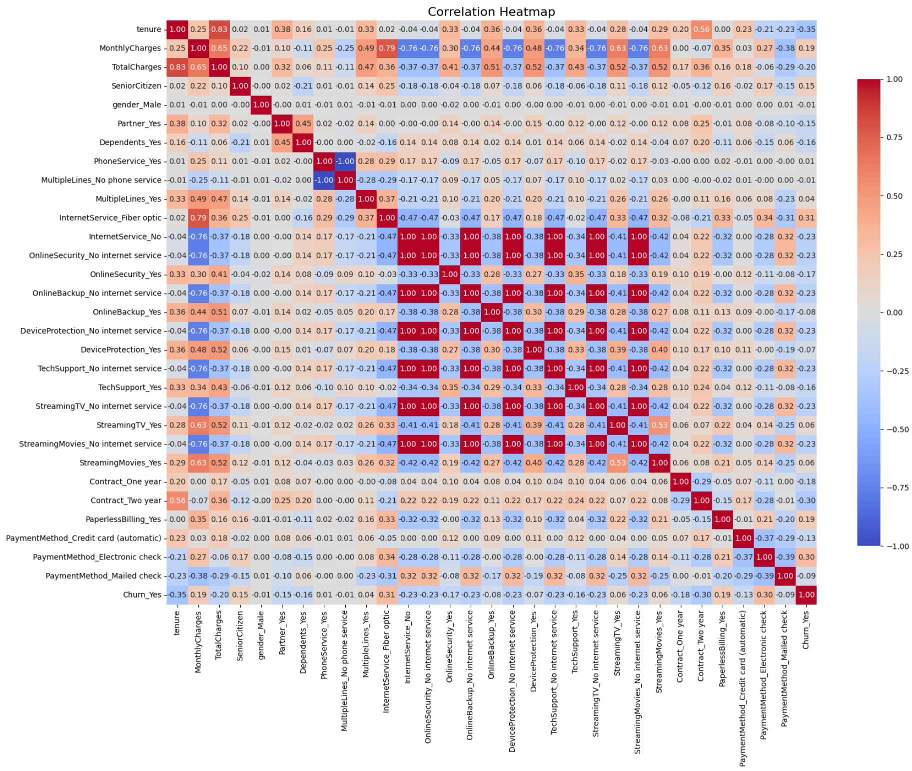
Box Plot Observations:



From the box plot, we observe no major extreme statistical outliers in the dataset. This was also evident in the histograms, as there were no significant gaps or extreme values. Tenure shows a broad spread, with the median around the mid-range. MonthlyCharges has a moderate spread, reflecting different service plans. TotalCharges shows the largest spread because it accumulates over time (tenure × monthly cost), which explains its longer whiskers.

Analysis of Features

Heat Map Observations:



The above heatmap describes the correlation between all the features.

Correlation with Churn (Churn_Yes):

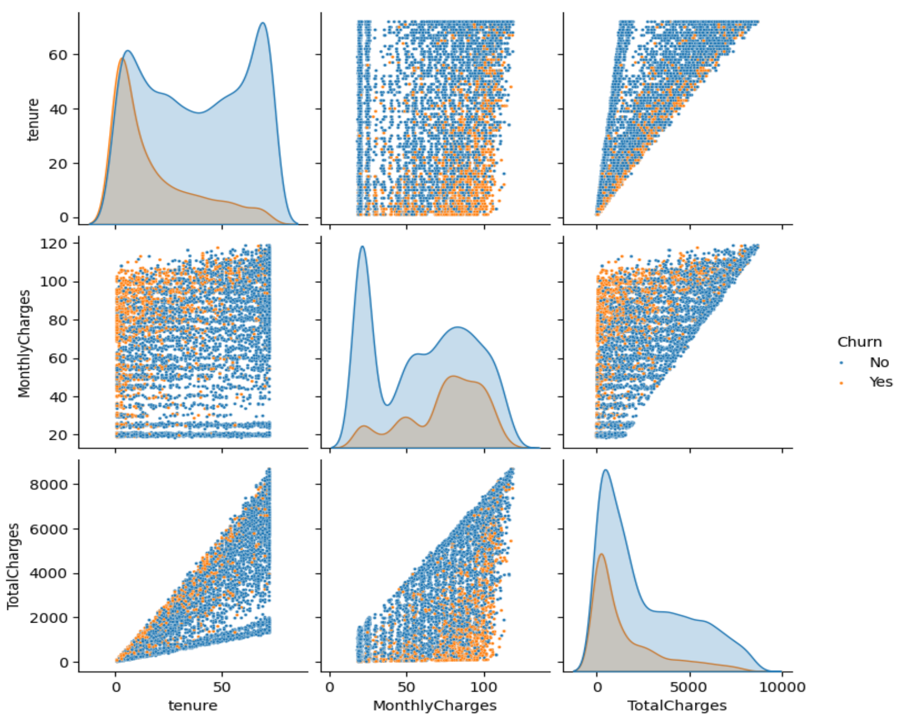
- Tenure: Strong negative correlation. Customers who stay longer are less likely to churn.
- Monthly Charges: Weak positive correlation. Higher monthly charges may contribute to churn.
- Total Charges: Negative correlation. Customers with lower total charges are more likely to churn, which aligns with its relationship with tenure.
- Senior Citizen: Weak positive correlation. Possibly due to lifestyle changes.
- Gender: Extremely weak correlation.
- Phone Service / Multiple Lines: Extremely weak correlation with churn.

- Auxiliary services (Device Protection, Streaming TV, Streaming Movies): Very weak correlations.
- InternetService_Fiber optic: Strong positive correlation with churn.
- PaperlessBilling and Electronic Check: Strong positive correlations with churn, especially electronic check.
- No Internet Service and Contract features: Strong negative correlation with churn, reinforcing the Fiber Optic observation.

Correlation Between Features:

- The strongest positive correlation is between Tenure and TotalCharges, which is self-evident.
- High MonthlyCharges correlate positively with Fiber Optic service.
- Fiber Optic service correlates positively with MultipleLines, PaperlessBilling, and Electronic Check payment methods, which is interesting form service perspective.

Cross Analysis of Numerical Features:

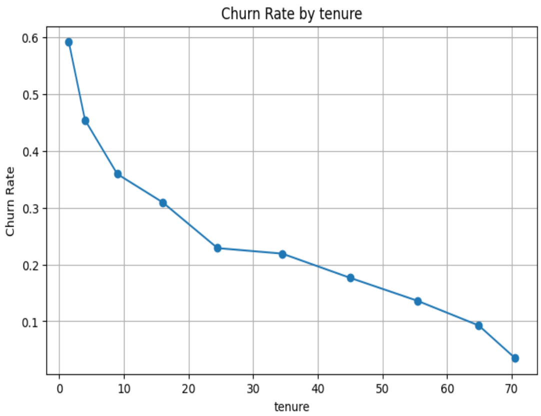


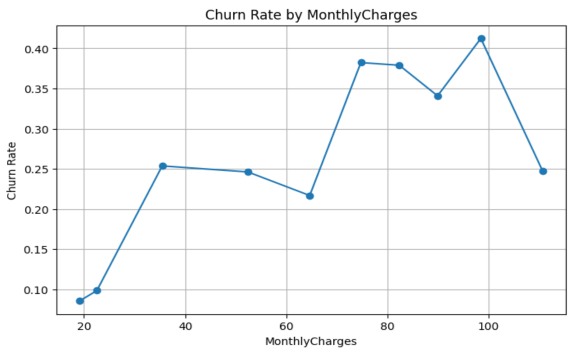
Each point in the pair plot represents a datapoint in selected dataset. We observe:

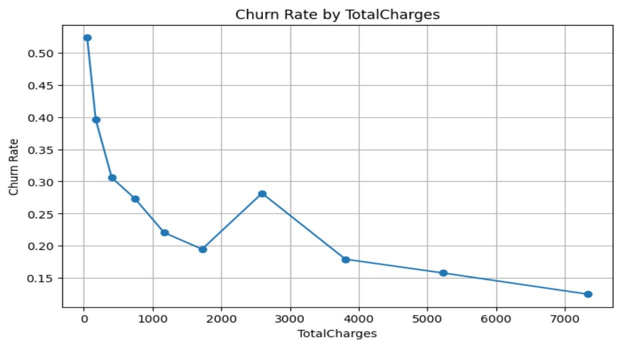
- Cluster of customers with low tenure and high monthly charges churned.
- The clustering between Tenure and Total Charges is not as clearly defined as that of the previous one.
- The clustering is well defined between Monthly Charges and Total Charges which logically indicates Tenure.

Model Selection

Churn problems are generally non-linear, our analysis suggested that tenure has a strong linear relationship with churn. However, since TotalCharges and some other features show non-linear patterns, we decided to compare Logistic Regression (linear model) and Random Forest (non-linear tree-based model). This allows us to compare linear and non-linear approaches fairly.







Model Training for Logistic Regression:

```
C_candidates = [0.01, 0.1, 1, 10]
penalty_candidates = ["l1", "l2"]
class_weight_candidates = [None, "balanced"]
```

- c_candidates → Strength of regularization
- penalty (l1, l2) → Features that were tuned during training. This changes how the model penalizes coefficient.
- class_weight → How heavily to penalize missed classification.
- l1 performs feature selection and removes weak predictors and is best used when you have many irrelevant features and high dimensional data.
- l2 keeps all features and reduces variance and is best used when you have many strongly correlated features and you think most of your features will influence the target.
- class_weight: When there is no class weight, misclassifying a no churn vs a yes churn carries the same weight: meaning they are as important to each other and when a balanced class weight, penalties for missed classification are weighted differently.

All permutations of these parameters were tested and arrived at the following results:

Best validation accuracy parameters are C = 1, Penalty is L2, Class_weight = None

Which yielded:

- 'training_accuracy': 0.8057700121901666
- 'validation_accuracy': 0.8085308056872038
- 'test_accuracy': 0.8056872037914692

Least overfit parameters are C = 0.01, Penalty is L2, Class_weight = none

Which yielded:

- 'training_accuracy': 0.8019097927671678
- 'validation_accuracy': 0.8009478672985783
- 'test_accuracy': 0.795260663507109

There is more information about different comparisons of these parameter sets in model_1_logreg_analysis_2026-03-02_01-16-55.txt. Of these two options, the first one seems to be the better model as the accuracy is higher without compromising very much on the fit.

Top 20 Feature importance of Best Validation Accuracy model:

Feature	**Coefficient**	**Abs_Coefficient**
tenure	-1.456742	1.456742
Contract_Two year	-1.326671	1.326671
InternetService_Fiber optic	0.943883	0.943883
TotalCharges	0.692233	0.692233
Contract_One year	-0.646919	0.646919
PhoneService_Yes	-0.497306	0.497306
OnlineSecurity_Yes	-0.402038	0.402038
PaperlessBilling_Yes	0.322583	0.322583
StreamingMovies_Yes	0.321465	0.321465
PaymentMethod_Electronic check	0.316522	0.316522
TechSupport_Yes	-0.311407	0.311407
MultipleLines_Yes	0.249137	0.249137
StreamingTV_Yes	0.202573	0.202573
SeniorCitizen_1	0.193747	0.193747
MonthlyCharges	-0.183243	0.183243

OnlineBackup_Yes	-0.166409	0.166409
Dependents_Yes	-0.135915	0.135915
DeviceProtection_No internet service	-0.132475	0.132475
InternetService_No	-0.132475	0.132475
OnlineSecurity_No internet service	-0.132475	0.132475

Effect of Preprocessing (Logistic Regression):

Standard Scaler: The Standard scaler puts all the data in the same scale. This is important because the numerical values of TotalCharges are much higher than the other numerical values. If we did not scale, these large numbers would skew the results putting even greater importance than warranted on TotalCharges (scikit-learn.org, n.d., 2025).

One Hot encoding: is extremely important for our model and dataset. If we did not use One Hot encoding and instead assigned each original categorical feature a number for their value, models would treat those values as a sort of scale. Such as 0: low, 1: medium, 2: large (geeksforgeeks, 2024). Our categorical data does not have this aspect. For example, if we did not use One Hot encoding, for Internet Service, the encoding would be something like this:

0: Fiber Optic

1: DSL

2: No Internet Service

And the models would assign some sort of relationship between these features based on their value, which does not reflect reality.

Model Training for Random Forest:

```
n_estimator_candidates = [100, 200, 300]
max_depth_candidates = [5, 10, 15, 20]
min_samples_split_candidates = [2, 5, 10]
```

- n_estimator_candidates → the number of trees to be used in the forest. More trees reduce overfitting and underfitting in general but is computationally expensive.
- max_depth_candidates → how far down a decision tree to traverse. High values lead to more overfitting, but less underfitting.
- min_samples_split_candidates → the minimum number of samples required to split an internal node. Higher values generally lead to less overfitting, but more underfitting.

All permutations of these parameters were tested and arrived at the following results:

Best Validation Accuracy parameters are n_estimators = 300, Max_depth = 15, min_samples_split = 5

Which yielded:

- 'training_accuracy': 0.9396586753352296
- 'validation_accuracy': 0.8037914691943128
- 'test_accuracy': 0.7914691943127962

Least Overfit parameters are n_estimators = 100, Max_depth = 5, min_samples_split = 5

Which yielded:

- 'training_accuracy': 0.8017066233238521
- 'validation_accuracy': 0.7867298578199052
- 'test_accuracy': 0.7838862559241706

There is more information about different comparisons of these parameter sets in model_2_rf_analysis_2026-03-02_16-49-07.txt. Unlike our results for Logistical Regression, the Best Validation Accuracy model had quite high overfitting. The overfitting seems to be high enough to warrant selecting a less accurate validation for one that is more fit to the data. The Least Overfit model does not have that much lower validation accuracy, so this is the parameter set for the model we have selected.

Top 20 Feature importance of Best Validation Accuracy model:

Feature	**Coefficient**	**Abs_Coefficient**
Tenure	0.246086	0.246086
TotalCharges	0.114873	0.114873
InternetService_Fiber optic	0.110715	0.110715
Contract_Two year	0.107701	0.107701
PaymentMethod_Electronic check	0.091711	0.091711
MonthlyCharges	0.054838	0.054838
OnlineSecurity_Yes	0.034814	0.034814
Contract_One year	0.032321	0.032321
TechSupport_No internet service	0.024898	0.024898
StreamingTV_No internet service	0.024399	0.024399
StreamingMovies_No internet service	0.021268	0.021268
OnlineSecurity_No internet service	0.020051	0.020051
InternetService_No	0.018117	0.018117
TechSupport_Yes	0.017084	0.017084
OnlineBackup_No internet service	0.01581	0.01581
DeviceProtection_No internet service	0.013561	0.013561
PaperlessBilling_Yes	0.006335	0.006335
SeniorCitizen_1	0.005956	0.005956

OnlineBackup_Yes	0.005863	0.005863
Partner_Yes	0.004927	0.004927

Effect of Preprocessing (Random Forest):

MinMax Scaler: Forest model is tree based and not based on the distance where features influence one another (Pramer, 2022). Therefore, a scaler does not need to be used with this model. Using one, however, should not have an adverse effect on the model itself. Either way, we tested both on the same data and the results were the same.

One Hot encoding: In general, one-hot encoding is not advised for Random Forest when the categorical features have high cardinality (Zabardast, 2024). Our dataset, however, has a maximum cardinality of 3, so we should not have this issue. Also, using One Hot encoding for Random Forest when there is low cardinality can improve the model ("Handling Categorical Variables," 2024).

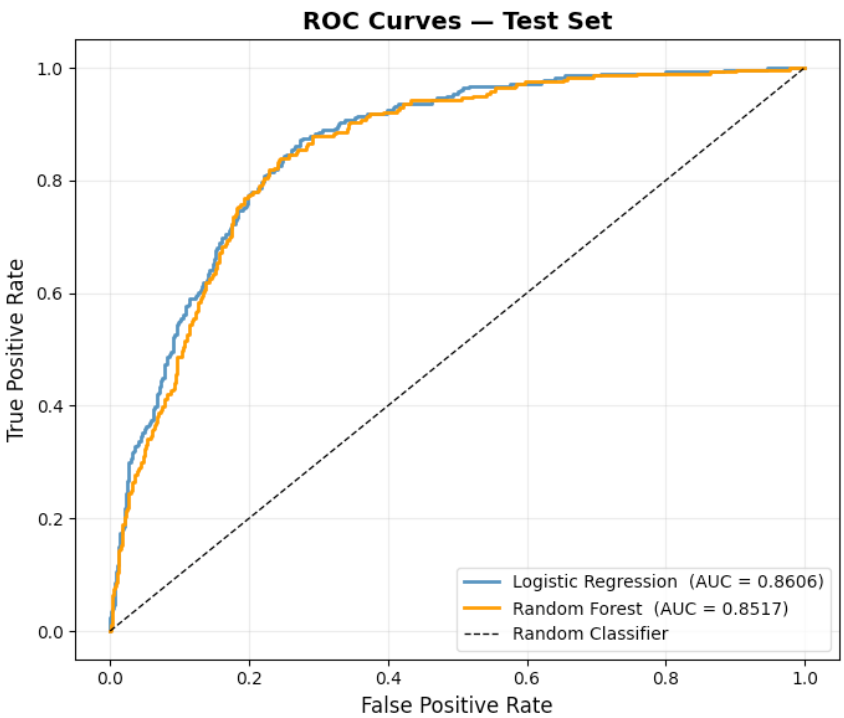
Compare of Logistical Regression and Random Forest:

Of the two models tested, Logistical Regression had higher accuracy with less overfitting in general.

Note that the fit of the best fitting model for Random Forest is still less well fit than the highest validation accuracy model for Logistical Regression. Additionally, the Highest Validation Accuracy for Logistical Regression also had higher accuracy than the Highest Validation Accuracy for Random Forest, while also being better fit than the Best Fit model for Random Forest.

With that in mind, Logistical Regression seems to be the superior choice for this problem overall. This is very interesting because churn datasets often contain more non-linear data and perform better with Tree-based models with Normalization. This could be because the highest correlation data (tenure) to our target is more linear, combined with the fact that this dataset has very few numerical features.

Metric	**Logistic Regression**	**Random Forest**
Training Accuracy (best val. model)	0.8058	0.9868
Validation Accuracy	0.8085	0.7982
Test Accuracy	0.8057	0.7944
Overfitting Risk	Low (train ≈ val)	High at max_depth=20
Best Least-Overfit Val Accuracy	0.8009	0.7905
Top Feature	tenure (coeff -1.457)	tenure (importance 38%)
2nd Feature	Contract_Two year	InternetService_Fiber optic
Interpretability	High	Moderate



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